

Report: Design, Fabrication and Analysis of Mach-Zehnder Interferometer in Silicon Photonics

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Abstract—This report presents the design and simulation of Mach-Zehnder interferometers (MZIs) optimized for silicon-on-insulator (SOI) fabrication. The transmission spectra were simulated to investigate the impacts of path length difference (ΔL), waveguide width, and silicon layer thickness on the device's optical properties. Finally, experimental data were compared against the simulation results to validate both the structural design and the fabrication process tolerances.

Index Terms—Mach-Zehnder interferometer, silicon photonics.

I. INTRODUCTION

The Mach-Zehnder interferometer (MZI) serves as a fundamental tool for measuring relative phase shifts between two coherent light beams. Utilizing the MZI architecture, various optoelectronic modulators, including electro-optic and thermo-optic variants, have been developed. This paper introduces a specific MZI design and evaluates its performance through detailed simulation results.

II. THEORY

Let us consider a symmetrical Y-junction splitter. An incident beam with an input intensity I_i and corresponding electric field E_i is divided equally between two arms. Consequently, the intensity and electric field in each branch are defined as $I_{1,2} = I_i/2$ and $E_{1,2} = E_i/\sqrt{2}$, respectively. Regarding the optical combiner, a single-branch input is distributed equally between the fundamental waveguide mode and higher-order radiation modes. Therefore, the resulting output field from a single arm is $E_{o,n} = E_n/\sqrt{2}$. When both arms are considered, the total output field E_o results from the vector superposition of the input fields, scaled by a factor of $1/\sqrt{2}$, yielding $E_o = (E_1 + E_2)/\sqrt{2}$. For a monochromatic plane wave, the electric field is represented by $E = E_0 e^{i(\omega t - \beta z)}$, where $\beta = 2\pi n/\lambda$ is the propagation constant and n represents the refractive index of the medium.

Assuming a lossless regime, the fields in the two arms can be expressed as

$$E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1} \quad (1)$$

$$E_{o2} = E_2 e^{-i\beta L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2} \quad (2)$$

By substituting these into the expression for the total output field, we obtain:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta L_1} + e^{-i\beta L_2}) \quad (3)$$

The resulting output intensity, $I_o \propto |E_o|^2$, can then be derived as follows:

$$I_o = \frac{I_i}{4} |e^{-i\beta L_1} + e^{-i\beta L_2}|^2 = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (4)$$

III. MODELLING AND SIMULATION

A. WAVEGUIDE

The proposed design utilizes a strip waveguide with a cross-section of 500 nm width and 220 nm height. All devices are designed for TE mode propagation, with the polarization in the plane of the chip. The electric field intensity profiles for the fundamental quasi-TE₀ and quasi-TM₀ modes are simulated using Lumerical MODE Solutions and presented in Figures 1 and 2, respectively.

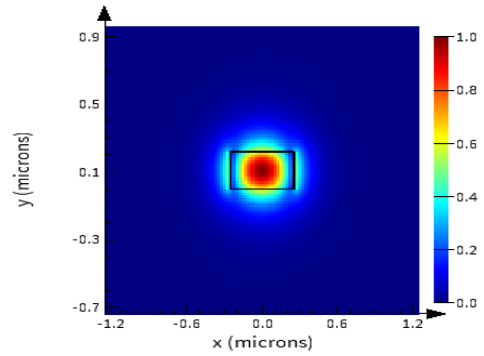


Fig. 1. E_x component of electric field intensity for TE₀ mode for 1550nm.

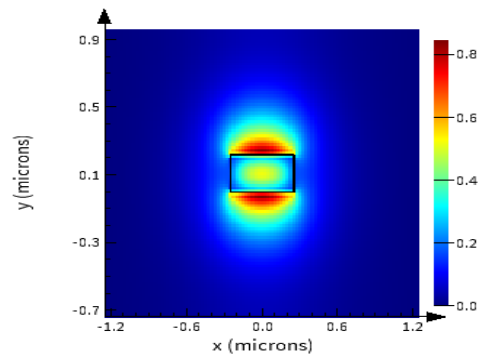


Fig. 2. E_y component of electric field intensity for TM₀ mode for 1550nm.

Furthermore, the spectral dependence of the effective index (n_{eff}) and the group index (n_g) for TE mode are plotted in Figures 3 and 4, respectively. As observed, the effective index decreases as the wavelength increases, whereas the group index exhibits an upward trend over the same spectral range.

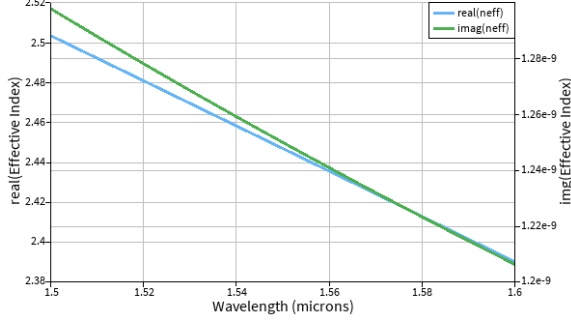


Fig. 3. Effective index of the waveguide versus wavelength.

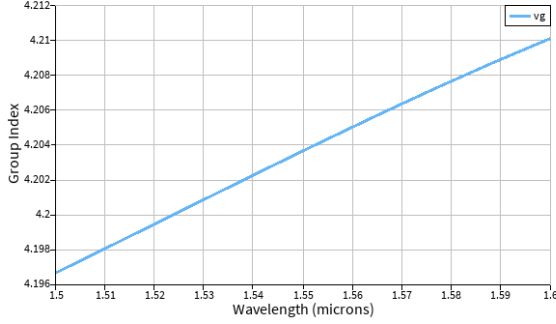


Fig. 4. Group index of the waveguide versus wavelength.

Using these data for n_{eff} and MATLAB fitting, a compact model was derived to characterize the waveguide's dispersion, yielding the following second-order polynomial fit:

$$n_{eff}(\lambda) = 2.4468 - 1.1333(\lambda - \lambda_0) - 0.0439(\lambda - \lambda_0)^2 \quad (5)$$

The corresponding graph is shown in Figure 5.

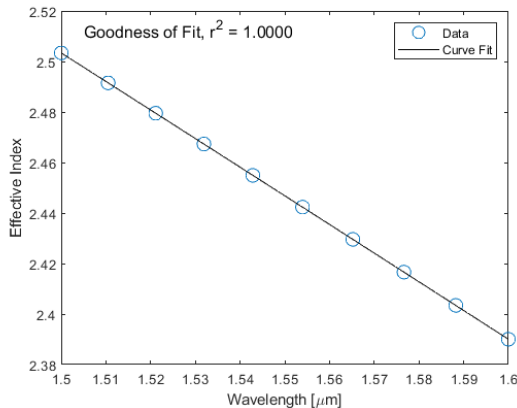


Fig. 5. Compact model for the waveguide mode TE₀.

B. MACH-ZEHNDER INTERFEROMETER

For a lossless Mach-Zehnder interferometer, the transfer function as a function of wavelength (λ) is expressed as :

$$T_{MZI}(\lambda) = \frac{I_{out}}{I_{in}} = \frac{1}{2}[1 + \cos(\beta\Delta L)] \quad (6)$$

where $\Delta L = L_1 - L_2$ is the path length difference and $\beta = \frac{2\pi}{\lambda}n_{eff}(\lambda)$ is the propagation constant, assuming $\beta_1 = \beta_2$.

The spectral distance between the peaks of the MZI transfer function, known as the free spectral range (FSR), is given by:

$$FSR = \lambda^2 / (n_g \Delta L) \quad (7)$$

where the group index, n_g , accounts for the dispersion of the waveguide and is defined as:

$$n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \quad (8)$$

Table 1 lists the various path length differences ΔL for the designed MZI devices alongside the expected FSR values for each device, calculated for TE₀ mode at $\lambda = 1550nm$ and $n_g = 4.2$.

TABLE I
FSR WITH DIFFERENT ΔL .

$\Delta L(\mu m)$	$FSR(nm)$
25	22.88
50	11.44
100	5.72
200	2.86
400	1.43
600	0.95

Next, we simulate the transmission spectrum of the MZI using Lumerical INTERCONNECT. We construct an interferometer circuit using Y-branch S-parameter elements from the course device library, and we add the TE-mode waveguide elements simulated by Lumerical MODE in the previous section. Figure 6 shows the simulation results for two devices with path length differences ΔL of $200\mu m$ and $100\mu m$.

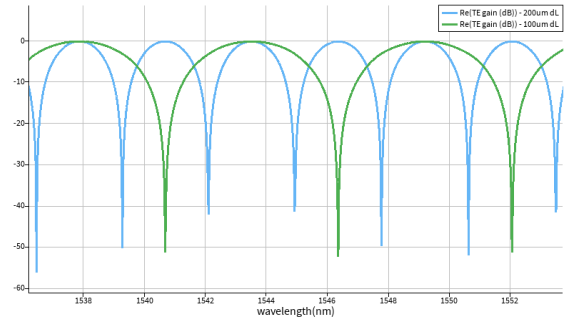


Fig. 6. Spectrum of MZI for two ΔL of $200\mu m$ and $100\mu m$.

Then, we simulate the transmission spectrum by taking into account the insertion loss from two grating couplers, using the data provided from the course device library. The resulting spectra are presented in Figure 7.

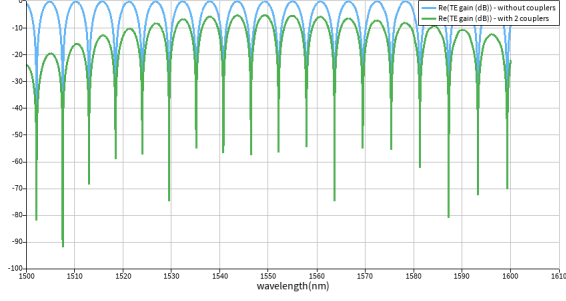


Fig. 7. Spectrum of MZI of $\Delta L = 100 \mu\text{m}$ with and without couplers.

Figure 8 shows insertion loss from two grading couplers calculated using Lumerical INTERCONNECT.

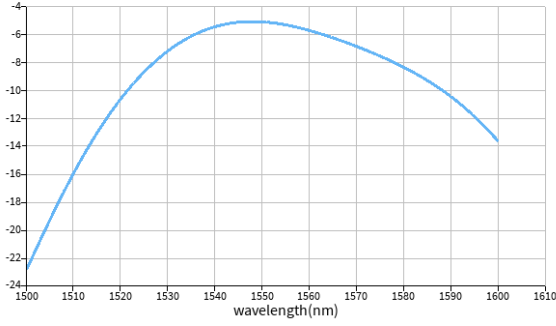


Fig. 8. Insertion loss of two grading couplers.

IV. LAYOUT

The design was developed using KLayout software. The final layout is presented in Figure 9, which contains nine devices: one de-embedded device, seven TE-mode 1x1 MZIs (single-output) with different ΔL , and two TE-mode 1x2 MZIs (double-output) with the same $\Delta L = 200 \mu\text{m}$. The purpose of designing devices with identical ΔL is to study manufacturing variability within a single chip. The details of these devices are presented in Table II.

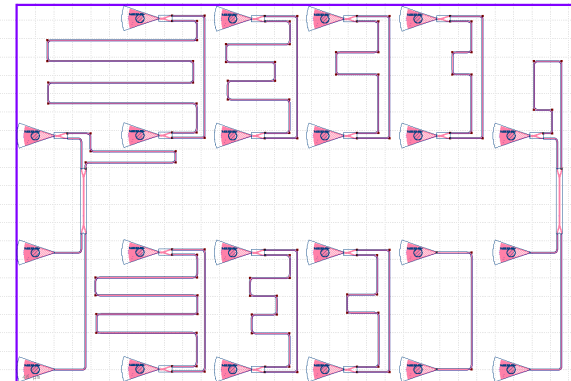


Fig. 9. Final version of the designed devices.

TABLE II
LIST OF FABRICATED DEVICES.

Device label	Device description	$\Delta L (\mu\text{m})$
REF	de-embedded	n/a
MZI1	balanced TE mode 1x1 MZI	0
MZI2	unbalanced TE mode 1x1 MZI	25
MZI3	unbalanced TE mode 1x1 MZI	50
MZI4	unbalanced TE mode 1x1 MZI	100
MZI5	unbalanced TE mode 1x1 MZI	200
MZI6	unbalanced TE mode 1x1 MZI	400
MZI7	unbalanced TE mode 1x1 MZI	600
MZI8A	unbalanced TE mode 1x2 MZI	200
MZI8B	unbalanced TE mode 1x2 MZI	200

VI. FABRICATION

A. FABRICATION AND MEASUREMENT DESCRIPTION

The fabricated is performed at Washington Nanofabrication Facility. The devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80°C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 °C, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH_4) flow of 13.0 sccm, nitrous oxide (N_2O) flow of 1000.0 sccm, high-purity nitrogen (N_2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon pieces on all sides to aid uniformity.

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

B. MANUFACTURING CHALLENGES

Fabrication processes introduce numerous sources of variation that alter structural parameters, thereby impacting device performance. For robust design, it is essential to account for deviations caused by lithographic smoothing, fluctuations in resist properties (e.g., thickness), exposure conditions, development time, chemical variations, and etching environments. According to data from the Washington University Nanofabrication Facility, the waveguide width typically varies between 470 and 510 nm. Additionally, the silicon layer thickness of the wafer deviates from the nominal 220 nm, ranging from 215.3 to 223.1 nm. Based on these structural bounds, a corner analysis was performed using Ansys Lumerical MODE to calculate the group index (n_g) at a wavelength of 1550 nm. This analysis evaluated the nominal as-designed case (500 nm width and 220 nm silicon thickness) alongside four extreme corner cases. Subsequently, the FSR for unbalanced MZI with $\Delta L = 200\mu\text{m}$ for each scenario was determined via the analytical formula given in Eq. (7) dependent on the calculated n_g values.

TABLE III
CORNER ANALYSIS.

Waveguide width (nm)	Silicon thickness (nm)	Group index n_g	FSR(nm) for $\Delta L = 200\mu\text{m}$
470	223.1	4.262	2.818
470	215.3	4.253	2.824
500 (nominal)	220 (nominal)	4.203	2.858
510	223.1	4.188	2.868
510	215.3	4.179	2.874

VI. EXPERIMENTAL RESULTS, ANALYSIS AND DISCUSSION

Experimental data were provided exclusively for devices 'REF', 'MZI1', 'MZI2', 'MZI3', and 'MZI8B', measured at temperatures of 25 °C and 30 °C. Due to inconsistencies observed in the 25 °C dataset, the analysis focuses solely on the 30 °C data points for these specified devices.

Figure 10 illustrates the raw data for the three 1x1 MZI devices. As expected, the balanced MZI1 device exhibits no oscillations, while the unbalanced MZI2 and MZI3 devices display spectral variations. Consequently, the FSR decreases proportionally with an increasing path length difference (ΔL).

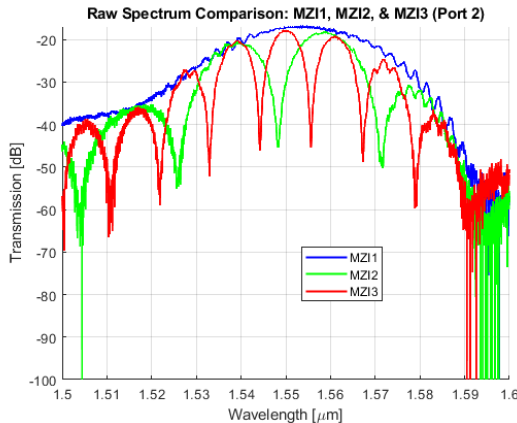


Fig. 10. Raw data for MZI1, MZI2, MZI3 (Port 2).

As expected, the balanced MZI1 device exhibited no oscillations, while the unbalanced MZI2 and MZI3 devices displayed clear spectral variations. Consequently, the FSR decreased proportionally with an increasing path length difference (ΔL).

To eliminate the spectral effects of the grating couplers from the MZI transmission spectra, a loopback structure consisting of two grating couplers connected by a waveguide path was utilized. Figure 11 shows the raw experimental data for the loopback structure alongside its mathematical fit. This baseline coupler loss was subsequently subtracted from the raw transmission spectra of the MZIs, yielding the calibrated transmission spectra shown in Figure 12. This normalization process effectively decouples the insertion loss of the input/output ports from the internal propagation behavior of the interferometer, ensuring that subsequent calculations of the group index remain unaffected by external coupling variations.

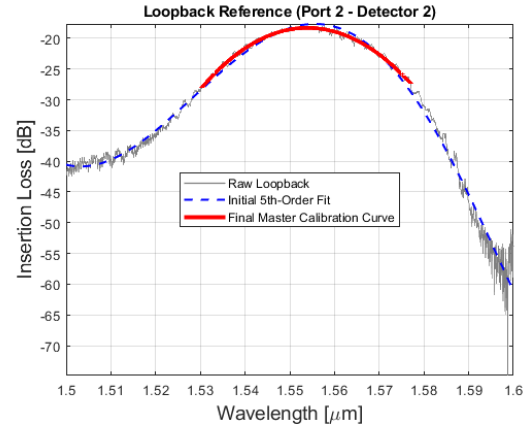


Fig. 11. Insertion loss corresponding to two grating couplers. Raw experimental data and fit.

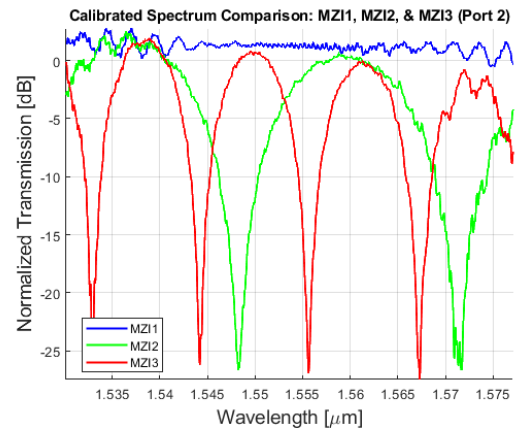


Fig. 12. Calibrated data for MZI1, MZI2, MZI3 (Port 2).

Figure 13 shows the raw data for the 1x2 MZI (double-output) device with $\Delta L = 200\mu\text{m}$, labeled as MZI8B, while Figure 14 depicts a zoomed-in view over the wavelength range of 1530–1570 nm. As expected, when light interferes constructively at output port 2 (resulting in maximum power), output port 3 experiences destructive interference with near-zero

power, and conversely. The anti-correlated spectral response reveals a high extinction ratio between the two ports across this standard telecommunications wavelength window. This stark out-of-phase relationship validates the device's functionality as a dual-output interferometric filter, making it a reliable candidate for extracting precise optical parameters.

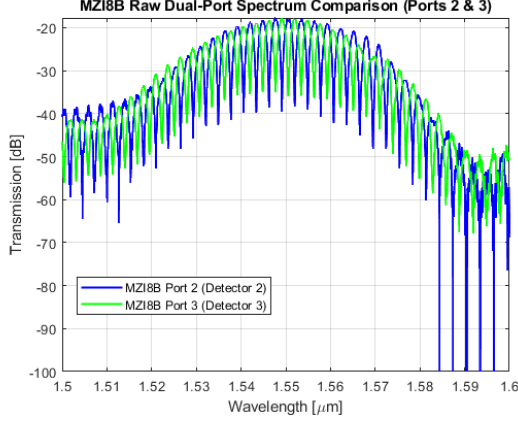


Fig. 13. Row data for MZI8B Port 2 and Port 3.

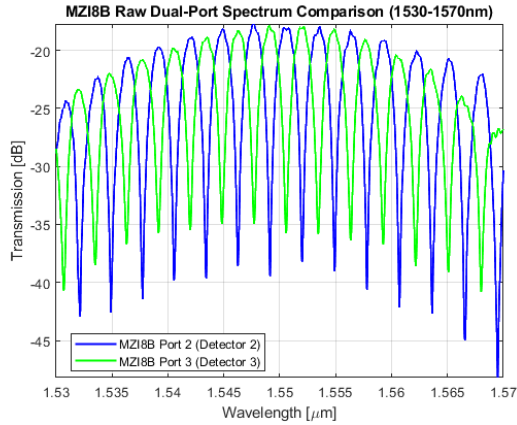


Fig. 14. MZI8B (Port 2 and Port 3) raw data in the range of wavelengths 1530-1570 nm.

A MATLAB script was used to identify the spectral minima, which can be precisely located due to their sharp profiles on a decibel scale, minimizing the impact of experimental noise on the extraction process. The free spectral range (FSR) and the resulting group index (n_g) were subsequently determined from the spacing of these minima using Eq. (7) for the wavelength of 1550 nm. Figures 15 and 16 present the extracted parameters for ports 2 and 3 of the MZI8B device, respectively. Analyzing both ports independently serves to validate the extraction model against the out-of-phase behavior of the 1x2 splitter. This same method was applied to the data for the MZI3 device with $\Delta L = 50 \mu\text{m}$ shown in Figure 17. However, the analysis of the MZI2 device ($\Delta L = 25 \mu\text{m}$) yielded inconsistent results due to the limited number of interference minima present within the spectrum. The consolidated final calculations for n_g are presented in Table IV.

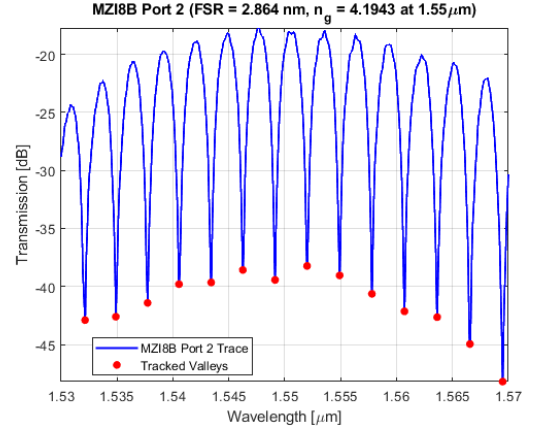


Fig. 15. MZI8B, Port 2 data treatment in the range of wavelength 1530-1570 nm.

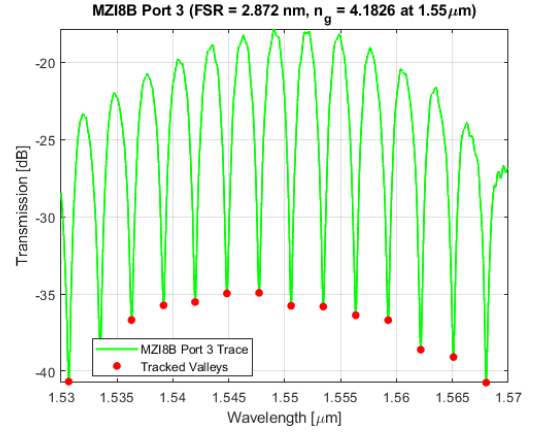


Fig. 16. MZI8B, Port 3 data treatment in the range of wavelength 1530-1570 nm.

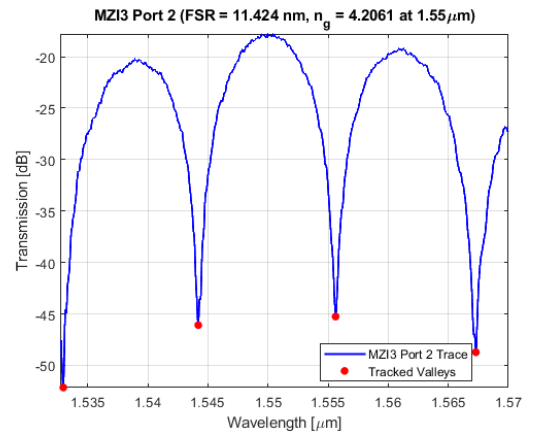


Fig. 17. MZI3, Port 2 data treatment in the range of wavelength 1530-1570 nm.

TABLE IV
SUMMARY OF EXPERIMENTAL DATA.

Device Type	Output (Port)	ΔL (μm)	FSR (nm)	n_g
MZI8B (double-output)	2	200	2.86	4.19
MZI8B (double-output)	3	200	2.87	4.18
MZI3 (single-output)	2	50	11.42	4.20

VII. CONCLUSION

In this report, several Mach-Zehnder interferometers (MZIs) with varying path length differences ΔL were successfully designed, fabricated, and characterized utilizing silicon strip waveguides on a silicon-on-insulator platform. Numerical simulations performed in Ansys Lumerical MODE established a robust corner analysis framework to account for real-world manufacturing tolerances, such as variations in waveguide dimensions and wafer thickness. Experimental spectral data obtained from the fabricated devices were systematically calibrated using an integrated loopback structure to isolate the core interferometric responses. By employing a precise MATLAB-based peak-detection script on the calibrated spectra, the free spectral range (FSR) and the corresponding group index (n_g) were successfully extracted at a target wavelength of 1550 nm. The close agreement between the simulated corner cases and the experimental results successfully validates both our design methodology and the fabrication process tolerances, offering a reliable approach for predictable parameter extraction in silicon photonic components.

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